Industrial Rope Access

Training Day 1

irata international Industrial Rope Access Trade Association MEMBER
SPRAT Society of Professional Rope Access Technicians
SKYTECH ACADEMY ROPE ACCESS TRAINING CENTER
OILFIELD | INDUSTRY | SAFETY | EXPERTISE

Classroom Rules and Expectations

- Safety First — Safety is the top priority at all times.
- Report all injuries to the instructor immediately, no matter how minor.
- Limit cell phone use. Please keep phones on silent during training sessions and store personal items in the designated area.
- You may store your lunch in the refrigerator; however, please do not consume anything that does not belong to you unless permission is given.
- Restrooms are available for use; please respect designated facilities.
- Remove all rope access equipment before entering office areas.
- Clean up after yourself and maintain a tidy training environment.
- Maintain a positive attitude and participate actively.
- Have fun and stay engaged.

Introduction — Rope Access Training

Instructor:

- Introduction of instructor
- Rope access experience:
- Training and instructional experience:

Students:

- Please introduce yourself:
- Your name and company:
- Your role at work
- Which training scheme and certification level you are attending
- Your previous rope access experience (if any)

Course Overview — Day 1

Day 1 focuses on the fundamentals of industrial rope access, including:

Safety principles and responsibilities

Basic equipment knowledge

Introduction to rope access systems

Operational awareness and safe practices

Why This Matters

This foundation is essential for all technicians

It ensures:

Safe working practices

Proper use of equipment

Readiness for practical rope access activities

Key Message

A strong understanding of fundamentals is critical before progressing to practical work at height.

Course Overview

Today's Agenda

Course overview and expectations

Introduction to safety principles and hazard awareness Overview of SPRAT standards and requirements Introduction to SPRAT and IRATA training schemes

Equipment & Safety

- Equipment familiarization

Inspection procedures (pre-use and formal checks)

Kitting up (correct harness and equipment setup)

Equipment pre-use inspection and function check

Roles & Worksite Awareness

Roles and responsibilities of rope access personnel

Housekeeping and site awareness

Evacuation routes and muster points

Introduction to Job Safety Analysis (JSA) for the work area

Administration

Required paperwork and documentation

Understanding training and certification requirements

End of Day

Post-training discussion and review

Equipment clean-up and storage

Final housekeeping of the work area

Key Message

Safety, preparation, and correct procedures are essential at every stage What you learn here forms the foundation for safe rope access work

Escape route

Training Objectives

Day 1 Learning Outcomes

By the end of Day 1, participants will be able to:

Understand the principles of industrial rope access systems Identify key components of Personal Protective Equipment (PPE) Recognize applicable industry standards (SPRAT & IRATA) Demonstrate safe equipment inspection practices Understand roles, responsibilities, and safety expectations

Key Message

These competencies form the foundation for safe and effective rope access work

Introduction to Rope Access

Industrial rope access is a safe and efficient method for working at height, allowing technicians to reach areas that are otherwise difficult or inaccessible. Its effectiveness comes from the strict application of proven safety principles.

Core Safety Principles

1. Redundant Systems

- Use of a primary working rope alongside an independent backup (safety) rope
- Provides fail-safe fall protection in case one system is compromised

2. Continuous Attachment (100% Tie-Off)

- Technicians are secured at all times while working at height
- Eliminates exposure to unprotected falls

3. Controlled Access and Descent

- Enables precise positioning and movement during work
- Maintains full control during ascent, descent, and task execution

Key Takeaway

Rope access is considered one of the safest height-access methods because it is built on:

- Redundancy — Backup systems prevent single-point failure
- Constant connection — Workers are always secured
- Controlled movement — Ensures stability and precision

When properly implemented, these principles make rope access a highly reliable and low-risk solution for working at height.

Training & Documentation Checklist

Mandatory Forms

Please complete the following:

All participants must complete the following documentation prior to attending training to ensure compliance, safety, and eligibility.

1. Mandatory Agreements

Liability Agreement

- Complete the Release of Liability and Assumption of Risk Agreement
- Print your name where indicated
- Initial each clause to confirm understanding and acceptance
- Sign and date the document

2. Required Supporting Documents

Participants must submit the following:

- Copy of government-issued identification (ID)
- SPRAT Evaluation Session Candidate Affidavit
- SPRAT Privacy Acknowledgement
- IRATA Candidate Disclaimer and Liability Release
- Certification of Physical Condition

3. Important Requirements

All documentation must be:

- Accurately completed
- Signed where required
- Submitted before participation in any practical training

Incomplete or missing documentation may result in delayed or denied participation.

Key Takeaway

- Proper documentation ensures:
- Compliance with training and industry standards
- Verification of participant eligibility

Training & Documentation Checklist

SPRAT Guidance & Technician Responsibilities

The Society of Professional Rope Access Technicians (SPRAT) provides essential standards and guidance to ensure safe and consistent rope access operations across the industry.

SPRAT Publications Include

SPRAT publishes key documents that define industry best practices:

- Safe Practices for Rope Access Work
- Certification Requirements for Rope Access Work
- Evaluation Guidelines
- Defined Terms and Industry Language

These publications establish the standards, expectations, and terminology used by rope access technicians worldwide.

Equipment Use & Care

SPRAT guidance covers the proper use, inspection, and maintenance of rope access equipment.

Technicians must always follow:

- Manufacturer instructions
- SPRAT recommended practices

Using equipment incorrectly or outside its intended purpose can compromise safety.

Technician Responsibility

Every technician is personally responsible for maintaining safe work practices.

You must:

- Ensure you have the latest information on equipment and procedures
- Regularly review manufacturer manuals and updates
- Verify correct equipment use prior to every task

If you are unsure:

- Consult your employer or supervisor
- Download and review the latest manufacturer instructions or official documentation

Training and Certification

SPRAT	Topic	IRATA
	Operations	
	Certification	
	Assessment Day	
	Training Required	
	Upgrade Requirement	
	Refresher	
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
	Certification	
	Assessment Day	
	Training Required	
	Upgrade Requirement	
	Refresher	
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
	Assessment Day	
	Training Required	
	Upgrade Requirement	
	Refresher	
	Supervision	
	Certification Cycle	

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Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
	Training Required	
	Upgrade Requirement	
	Refresher	
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
Yes, affidavit	Training Required	Yes, 32 Hours
	Upgrade Requirement	
	Refresher	
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
No, Recommended	Training Required	Yes, 32 Hours
500 hours and 6 months	Upgrade Requirement	1000 hours and 1 year
	Refresher	
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
No, Recommended	Training Required	Yes, 32 Hours
500 hours and 6 months	Upgrade Requirement	1000 hours and 1 year
Recommended	Refresher	8 hours if inactive for 6 months or more
	Supervision	
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
No, Recommended	Training Required	Yes32 Hours
500 hours and 6 months	Upgrade Requirement	1000 hours and 1 year
Recommended	Refresher	8 hours if inactive for 6 months or more
Level2 in limited circumstances	Supervision	Level3
	Certification Cycle	

Training and Certification

SPRAT	Topic	IRATA
Safe Practices for Rope Access Work	Operations	International Code of Practice
Certification Requirements for Rope Access Work	Certification	Training, Assessment, and Certification Scheme
Evaluation Requirements	Assessment Day	Training, Assessment, and Certification Scheme
No, Recommended	Training Required	Yes, 32 Hours
500 hours and 6 months	Upgrade Requirement	1000 hours and 1 year
Recommended	Refresher	8 hours if inactive for 6 months or more
Level 2 in limited circumstances	Supervision	Level 3
3 Years	Certification Cycle	3 Years

Training Safety Rules & Requirements

Safety First — Always

- Treat the training environment as a real worksite at all times
- Maintain a safety-first mindset in every activity

Personal Protective Equipment (PPE)

- Helmets are mandatory inside designated exclusion zones
- Gloves are recommended and available for purchase:

5. Leather gloves — approx. $25
6. General work gloves — approx. $10

Proper PPE use is essential to minimize risk and prevent injury.

Facilities & Equipment

First Aid Kit

- Located within the training center
- May be used when needed
- Inform the instructor of any items used

Drinking Water

- Readily available in the training center
- Stay hydrated throughout training

Worksite Procedures

- A Job Safety Analysis (JSA) must be:

13. Completed every morning
14. Signed by all participants before starting work

Participant Responsibilities

- Follow the JSA at all times
- Work in accordance with identified hazards and control measures

Key Takeaway

- Follow all rules
- Use equipment properly
- Stay alert and aware at all times
- Safe behavior is mandatory — no exceptions

Practical Exercises – Day 1

LEVEL 1	LEVEL 2	LEVEL 3
Kitting up	Horizontal Aid Climb	Level 2 exercises
Use of Equipment	Vertical Aid Climb	Pitch head Platform haul
Changeover	Rescue kits	Anchors pre -rigged to lower
Pass Deviation (S – Single) (I – Single, Double)	Lowering Systems	Rescue kits
Transfer	Pitch head / platform haul	Hauling and Lowering through knots
Knots (8,9)		

Time to Climb!

Industrial Rope Access

Training Day 2

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Working at Height Rules

Global Regulatory Overview

Working at height is governed by different regulatory bodies worldwide, each with its own standards and requirements:

- United States: OSHA (supported by ANSI Z359 standards)
- United Kingdom: HSE (British Standards — BS)
- European Union: EN Standards
- Canada: CCOHS (CSA Standards)

While regulations differ by region, core safety principles remain similar, though specific requirements, procedures, and compliance details may vary significantly.

ANSI Standards — Practical Use (U.S. Focus)

Although ANSI standards are voluntary, they play a critical role in industry safety:

- OSHA frequently references ANSI standards, making them widely accepted as best practice
- Following ANSI is typically safer and more efficient than developing internal standards from scratch

ANSI Z359 Fall Protection System Includes:

- Z359.1 / Z359.2 — Program management requirements
- Z359.3 — Safety requirements for rescue systems
- Z359.11 — Performance requirements for full-body harnesses

Equipment & Compliance

- All fall protection equipment (e.g., harnesses, lanyards, connectors) must meet ANSI performance requirements
- Manufacturers are responsible for:

12. Testing equipment against ANSI standards
13. Ensuring compliance and proper labeling

Equipment must always be used in accordance with manufacturer instructions and applicable standards.

Key Takeaway

In the United States:

- Follow OSHA regulations
- Use ANSI-compliant equipment
- Always inspect equipment and check tags before use

Understanding and adhering to these standards ensures legal compliance, safety, and best practice in working at height.

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Workplace Safety Overview

OSHA & Workplace Safety Overview

1. What is OSHA?

- OSHA = Occupational Safety and Health Administration
- U.S. authority responsible for:

3. Developing safety regulations
4. Enforcing workplace safety laws

- Applies to all workplaces in the United States

2. OSHA Standards Structure

- 29 CFR 1910 □ General Industry
- 29 CFR 1926 □ Construction

These are the main regulatory frameworks for workplace safety.

3. Working at Height (WAH)

- Not always covered by a single rule
- Requirements are often:

10. Spread across multiple OSHA standards
11. Referenced indirectly

4. Role of ANSI Standards

- OSHA relies on ANSI (American National Standards Institute)
- Key standard:

14. ANSI Z359 — Fall Protection Code

Includes:

- Equipment (harnesses, lanyards)
- Systems (lifelines, anchors)
- Training requirements
- Rescue procedures

Workplace Safety Overview

5. How OSHA & ANSI Work Together

- OSHA = Legal requirement (mandatory)
- ANSI = Technical guidance (best practice)
- OSHA often references ANSI

ANSI becomes expected industry standard

Key Point

Following OSHA may require using ANSI-compliant equipment and procedures

ANSI provides the detailed "how-to" for compliance

Quick Summary

- Follow OSHA regulations (required)
- Apply ANSI standards (best practice)
- Ensure fall protection systems meet Z359 requirements

OSHA vs ANSI Compliance — Practical Understanding

1. Understanding the Difference

OSHA

- Legal authority in the U.S.
- Compliance is mandatory
- Defines required workplace safety rules

ANSI

- Develops consensus safety standards
- Voluntary by definition
- Provides detailed technical guidance

2. How It Works in Practice

- OSHA regulations often reference ANSI standards
- This makes ANSI the expected industry best practice

Workplace Safety Overview

3. Equipment Standards (ANSI Role)

ANSI standards define how fall protection equipment must perform:

- Strength and load capacity
- Design and construction requirements
- Testing and performance verification

Example: Harnesses must comply with ANSI Z359 requirements

4. Manufacturer Responsibilities

Manufacturers ensure compliance by:

- Testing equipment during design and production
- Using ANSI standards as the performance benchmark
- Certifying products meet or exceed required safety criteria

5. Practical Reality

Even though ANSI is labeled "voluntary":

- It is widely used across the industry
- It is referenced by OSHA regulations
- It is critical for equipment safety and reliability

Key Takeaway

- OSHA = What you must follow (law)
- ANSI = How you safely meet those requirements

Real compliance requires both OSHA regulations and ANSI standards working together

ANSI Z359 Fall Protection Standards

The ANSI Z359 Series provides technical standards for:

- Fall protection systems
- Equipment requirements
- Program management

Key Standards Breakdown

1. ANSI Z359.1 — Fall Protection Code

- Defines minimum guidelines for fall protection programs
- Covers:

6. Systems and components
7. Work processes
8. General fall protection requirements

2. ANSI Z359.2 — Managed Fall Protection Program

- Establishes requirements for employers to:

10. Develop and maintain a comprehensive fall protection program

- Includes:

12. Policies and procedures
13. Roles and responsibilities
14. Training requirements
15. Program implementation and management

3. ANSI Z359.3 — Rescue Systems

- Covers requirements for:

17. Assisted rescue systems
18. Self-rescue systems

4. ANSI Z359.11 — Full Body Harnesses

- Defines requirements for:

2. Design and construction
3. Performance and strength

Applies to full body harnesses used in:

- Personal fall arrest systems
- Rope access operations

Key Takeaway

- ANSI Z359 provides the technical foundation for fall protection safety
- Covers everything from program management to equipment performance
- Essential for ensuring safe work at height and regulatory compliance

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Definitions

Fall Protection — Definition & Understanding

Why Definitions Matter

Understanding key terms is essential.

Without clear definitions, safety concepts can be misinterpreted or meaningless in practice.

Definition: Fall Protection

- Fall Protection is a backup system designed for workers at height
- It is used when there is a risk of losing balance or falling

Purpose of Fall Protection

- To control a fall if it occurs
- To eliminate or reduce the risk of injury
- To ensure the worker remains safe while working at height

1. Fall Arrest

- A form of fall protection designed to safely stop a person who is already falling
- Used when a fall cannot be completely prevented
- Requires:

9. Full body harness
10. Energy-absorbing system
11. Secure anchor point

Purpose: Minimize injury by controlling and stopping the fall

2. Work Positioning

A system that allows a worker to be:

- Supported on a vertical surface (wall, tower, structure)
- Working hands-free

Uses:

- Harness or body belt
- Ropes, lanyards, and connectors

Purpose: Provide stable positioning to perform work safely

Definitions

3. Rope Access (Relation to Work Positioning)

- Rope access is a form of work positioning system
- Technicians:

3. Are supported by ropes
4. Maintain hands-free work capability
5. Use controlled movement for access and positioning

4.Fall Restraint

- A form of fall protection where the worker is:

7. Restrained from reaching a fall hazard

- Prevents the worker from:

9. Accessing edges or unsafe areas
10. Being exposed to a fall situation

Purpose: Eliminate the possibility of a fall before it can occur

5.Managed Fall Arrest Program

- A formal program developed by the employer to:

12. Protect employees exposed to fall hazards

- Includes:

14. Procedures and policies
15. Equipment selection and use
16. Training and competency requirements
17. Inspection and maintenance processes

Purpose: Ensure a systematic and controlled approach to fall protection

Key Takeaway

- Fall Arrest = Protection after a fall starts
- Work Positioning = Prevention and control while working
- Rope Access = Advanced work positioning system with controlled movement
- Fall Restraint = Prevents access to hazard (no fall possible)
- Managed Fall Arrest Program = Employer system to manage fall risks

Fall Arrest – ANSI Z359.2

Fall Arrest (ANSI Z359.2)

Fall arrest systems are designed to stop a worker after a fall has occurred, preventing contact with the ground or lower level.

In fall arrest:

- The worker has already fallen
- The harness and connecting system absorb energy and safely stop the fall

Important Reality

A fall arrest event can be:

- Physically demanding and uncomfortable
- Associated with high forces on the body
- Proper setup is critical to minimize injury

Fall Arrest Attachment Points

There are two primary harness connection points used for fall arrest:

- **Sternal (Front Chest D-Ring)**

Used for limited fall distances

Lower force limits

- **Dorsal (Back D-Ring – between shoulder blades)**

Primary point for traditional fall arrest systems

Allows greater fall distance and higher forces

Key Takeaway

Fall arrest = last line of defense

Always:

- Minimize fall distance
- Select the correct attachment point
- Ensure system is properly designed

Fall Arrest – ANSI Z359.2

Fall Arrest Requirements (ANSI Z359.2)

ANSI Z359.2 defines requirements for fall arrest systems, including limits based on connection points:

Sternal (Front) Connection

- Maximum fall distance: 2 ft
- Maximum arrest force (peak): 900 lbs (4 kN)
- Average arrest force: 450 lbs (2 kN)

Dorsal (Back) Connection

- Maximum fall distance: 6 ft
- Maximum arrest force (peak): 1,800 lbs (8 kN)
- Average arrest force: 900 lbs (4 kN)

Key Takeaway

- Sternal connection = lower forces, shorter falls
- Dorsal connection = higher limits, longer allowable fall distance

Fall Arrest – ANSI Z359.2

Fall Arrest Requirements (ANSI Z359.2)

Connection Point Comparison

Sternal (Front Connection)

- Maximum fall distance: 2 ft
- Maximum arrest force (peak): 900 lbs (4 kN)
- Average arrest force: 450 lbs (2 kN)

Dorsal (Back Connection)

- Maximum fall distance: 6 ft
- Maximum arrest force (peak): 1,800 lbs (8 kN)
- Average arrest force: 900 lbs (4 kN)

Side-by-Side Summary

Sternal Connection

- Shorter fall distance
- Lower forces on the body
- More controlled fall

Dorsal Connection

- Longer allowable fall distance
- Higher arrest forces
- Commonly used in general fall arrest

Key Takeaway

Sternal = Lower force + Short fall

Dorsal = Higher force + Longer fall

Selection depends on application, system design, and safety requirements

Fall Arrest

Rope Access Harness and Fall Arrest Use

A typical rope access harness is a one-piece full-body harness, such as models from manufacturers like Petzl (e.g., AVAO BOD).

When fitted and used correctly, this type of harness meets ANSI requirements for fall protection and can be safely used for fall arrest.

Approved Fall Arrest Connection Points

A properly designed rope access harness can be used for fall arrest from the following attachment points:

Sternal (Front) Attachment Point

- Used for rope access systems and limited fall arrest
- Must meet reduced fall distance and force limits

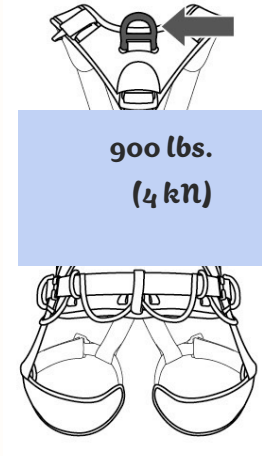
Dorsal (Back) D-Ring

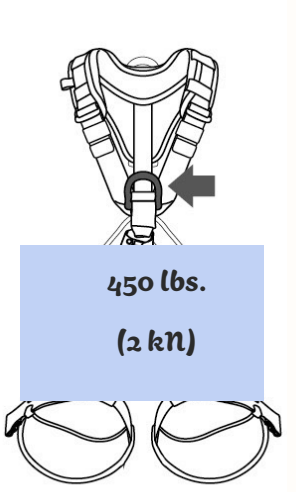
- Primary connection for traditional fall arrest systems Allows greater fall distance but with higher force limits

Ventral (Lower Front) Attachment Point

- Typically used for rope access, positioning, and suspension Not normally used for fall arrest unless specifically rated









Work Positioning - ANSI Z359.2

Work Positioning (ANSI Z359.2)

Work positioning systems are designed to hold a worker in place while preventing access to fall hazards.

In this system:

- The worker is restrained from reaching an edge
- It is used for support and stability, not fall arrest

Attachment points:

- Typically include side D-rings or ventral points, depending on the task A secondary fall protection system is required when exposure to a fall hazard exists

Equipment Requirements

D-Rings:

Minimum breaking strength 5,000 lbs (22.2 kN)

Buckles:

Minimum breaking strength 4,000 lbs (17.8 kN)

Key Takeaway

Work positioning = fall prevention, not fall arrest Worker must be kept away from the hazard A backup system is required if there is any fall risk

Work Positioning - ANSI Z359.2

Work Positioning Anchorage Requirements (ANSI Z359.2) Anchorages used for work positioning systems must meet the following requirements:

Strength Requirements

Certified Anchorage:

- Must support at least 2 x the foreseeable force applied to the system Designed and approved by a qualified person

Non-Certified Anchorage:

- Must support a minimum of 3,000 lbs (13.3 kN) per worker

System Use Requirements

- When using a segmented anchorage system:
- Only one worker may be attached per segment at a time Anchor points must be:
- Properly positioned to prevent reaching a fall hazard
- Compatible with the positioning system components

Key Takeaway

Work positioning anchorages have lower load requirements than fall arrest Each system must be used as designed and within its limits Always verify anchorage rating and configuration before use

Work Positioning – ANSI Z359.2

Rope Access Harness – Work Positioning (Quick Guide)

Harness Type

- Full-body rope access harness (e.g., Petzl AVAO BOD)
- Must meet ANSI requirements

Approved Work Positioning Attachment Points

- **Lateral (Side D-Rings)**

4. Primary work positioning points
5. Allow stable, hands-free work

Ventral (Front Lower Point)

6. Common in rope access systems
7. Provides central balance and control

Important Usage Rules

- Harness must be:

9. Properly fitted to the user
10. Used per manufacturer instructions

- System must be:

12. Compatible with anchor and rope setup

Not all attachment points are suitable for every task

Always use the correct designated point

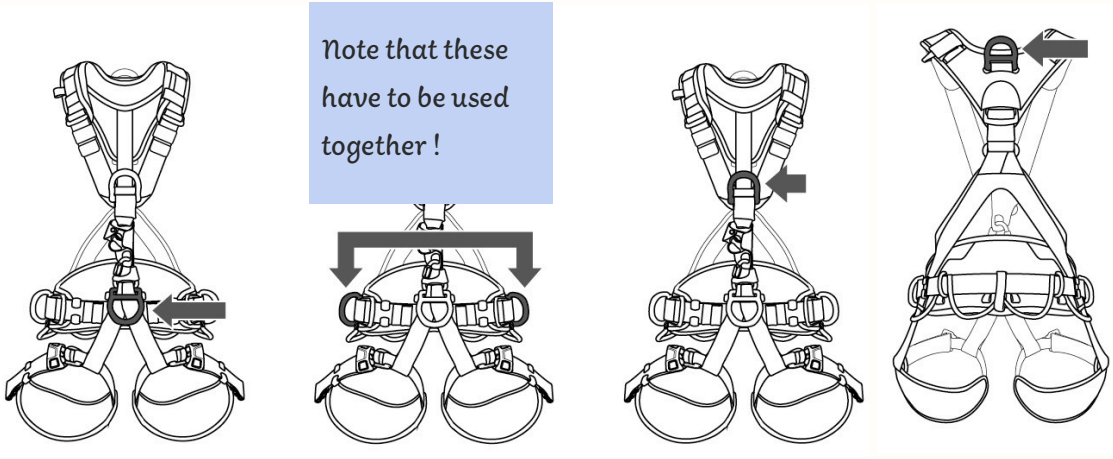
Pre-Use Check

- Fit adjusted correctly
- Connection point selected properly
- Equipment compatible with system
- No visible damage or defects

Key Takeaway

- Rope access harnesses support multiple functions
- Side and ventral points are used for work positioning
- Correct use ensures:

8. Stability
9. Safety
10. ANSI compliance



Fall Protection

Fall Protection Saves Lives

Falls from height are one of the leading causes of serious injury and fatalities in the workplace

Most incidents happen because of:

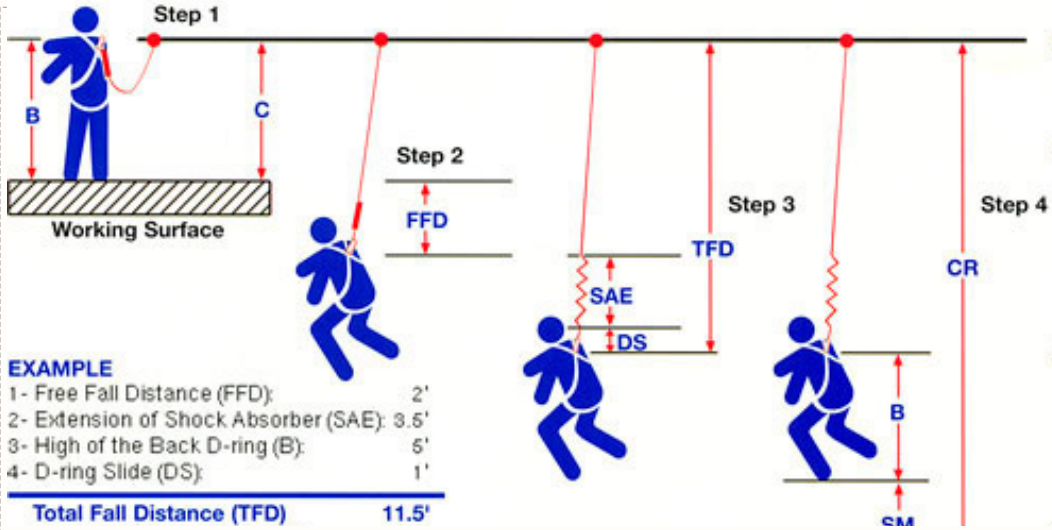
- Incorrect equipment use
- Poor anchor selection
- Lack of inspection or training

Your Life Depends on It

Fall protection is not optional Small mistakes at height can have serious or fatal consequences

Clearance Distance

CALCULATING - CLEARANCE REQUIREMENT FOR SHOCK ABSORBING LANYARD



EXAMPLE

1- Free Fall Distance (FFD):	2'
2- Extension of Shock Absorber (SAE):	3.5'
3- High of the Back D-ring (B):	5'
4- D-ring Slide (DS):	1'
Total Fall Distance (TFD)	**11.5'**

Step 1: Always choose an anchorage above the back D-ring whenever possible.

Step 2: Calculate the free fall distance (FFD) of the system by:
FFD = Length of lanyard (L)
+ Height of TL back D-ring (B)
- Height of anchorage (C)

Step 3: Calculate total fall distance (TFD)
TFD = FFD (Free fall distance)
+ SAE (Extension of shock absorber)
+ DS (D-ring slide)

Fall Factor

Fall Factor is an easy way for us to understand the severity of a fall, if it occurs.

Fall Factor ⟶ Severity of Fall

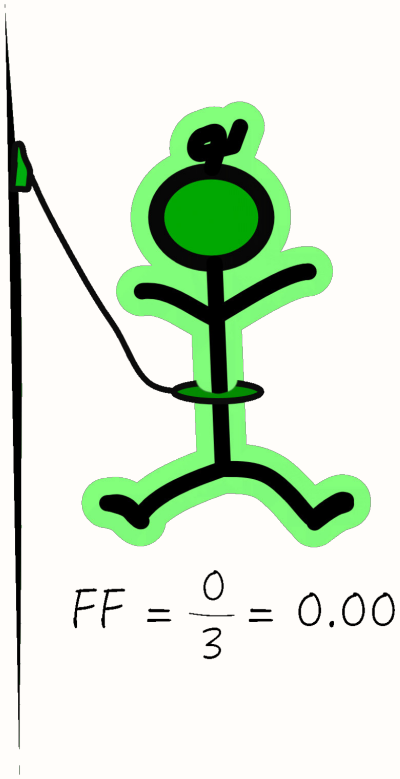
Calculated by dividing the distance of fall by the length of rope in the system.

$$\frac{\text{Length of Fall}}{\text{Length of Rope}}$$

So, imagine a worker using a 3' dynamic cowstail connected to his backup safety device ...

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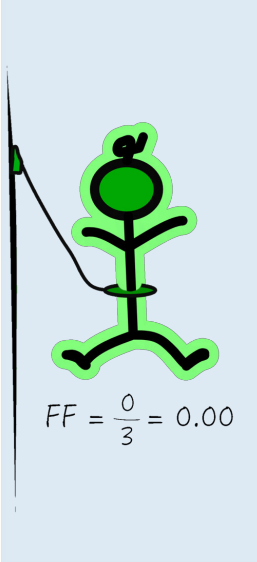
Fall Factor

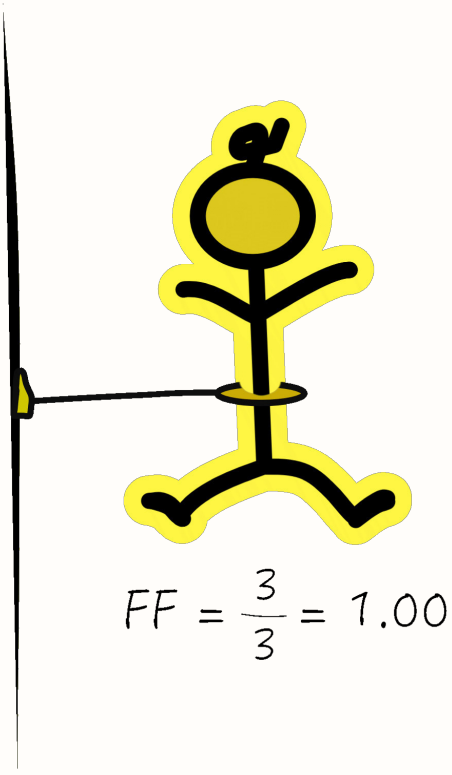


$$FF = \frac{0}{3} = 0.00$$

FF =	Fall	/	Rope
0	**0'**	**/**	**3'**

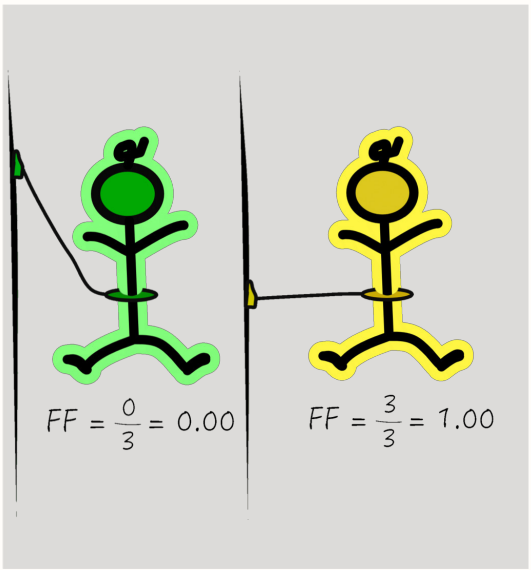
Fall Factor

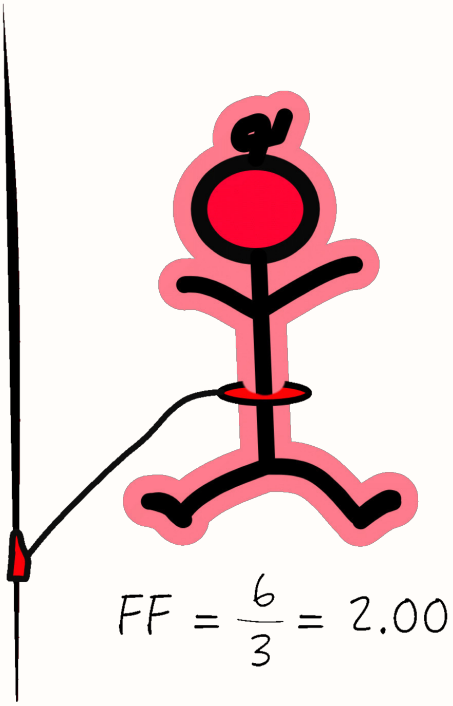




FF =	Fall	/	Rope
0	0'	/	3'
1	**3'**	**/**	**3'**

Fall Factor





FF =	Fall	/	Rope
0	0’	/	3’
1	3’	/	3’
2	6’	/	3’

Practical Exercises — Day2

LEVEL 1	LEVEL 2	LEVEL 3
Repeat Day 1 activities	Rigging Rope Systems	Use of Tensioned Lines
Pass short re-anchor (1)	Rope and Sling Protection	Complex Rescue Scenario
Pass long re-anchor(S)	Pre-rigged to Lower Systems	Work Scenario
Pass Knots	Retrievable Rigging	
Fixed aid climb	Load Sharing Anchors	
Mobile aid climb	Cross Haul	
Climbing with Fall Arrest Lanyards (1)	Tensioned Lines (1)	
	Rigging Fall Restraint Lines (1)	
Knots (alpine, stop, join)	Knots (Prusik, Munter)	

Time to Climb!

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Training Day 3

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Anchor Systems

Anchor Principles — Critical Safety Information

Fundamental Rule

No anchor = No safe system

Anchors are the foundation of all work at height.

Failure of an anchor results in total system failure.

Core Requirements

Anchors must:

- Be strong enough for the intended load
- Be the correct type for the task
- Be connected to a structurally sound anchorage

System Rules (Non-Negotiable)

- Always use TWO independent anchors
- Anchors must be:

6. Separate
7. Independent

Failure of one anchor must NOT affect the second

Positioning Rules

Anchors should be placed to:

- Minimize fall distance
- Reduce swing fall risk

Load Management

The system should:

- Distribute loads effectively
- Use load sharing when applicable
- Avoid overloading a single anchor point

Protection & Setup Rules

Anchors must be:

- Protected from:

2. Sharp edges
3. Heat
4. Abrasion
5. Chemicals

- Installed to:

7. Prevent movement
8. Avoid accidental disconnection

Failure Concept

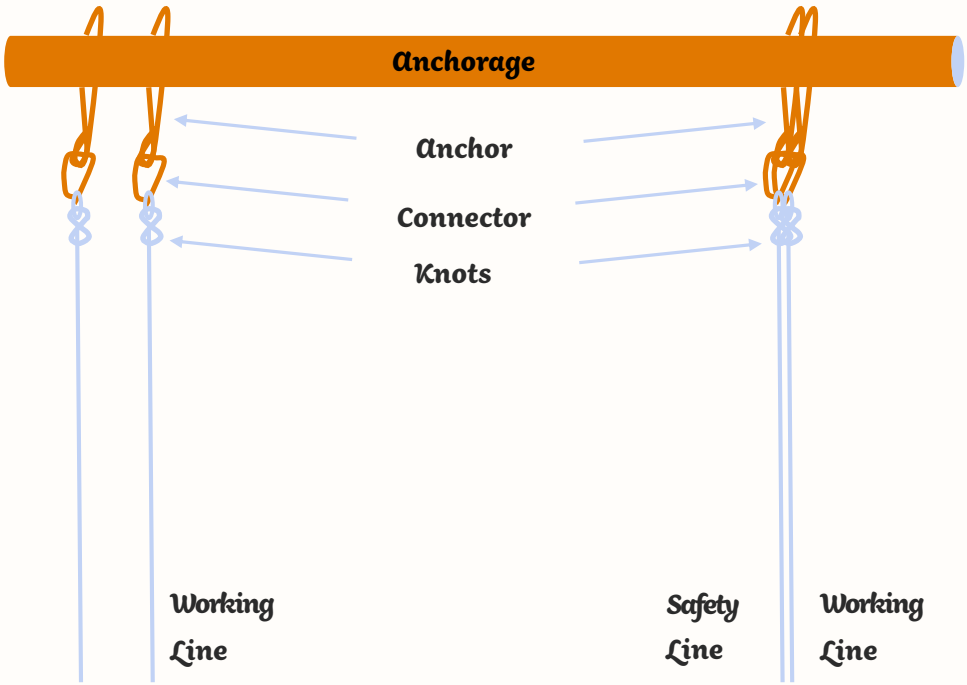
Anchor failure = Complete system failure

Key Takeaway

- Use two independent anchors
- Select and position anchors carefully
- Protect and manage loads properly

Safe rope access starts with proper anchor setup and control

Anchor Systems



Load sharing is achieved by placing the bight of the knot through both connectors

Load Angles

Anchor Angles & Load Distribution

What is the Included Angle?

The included angle is:

- The angle between two anchor lines (rope-to-rope)
- Measured on the inside of the system

How Angle Affects Load

As the included angle increases:

- Force on each anchor increases
- Load sharing becomes less efficient

Key Angle Rules

Preffered Zone

- ≤ 90°
- Best load distribution
- Lower force on each anchor

Critical Zone

- ≈ 120°
- Each anchor carries ~100% of the load

Danger Zone

- > 120°
- Forces increase rapidly and exponentially
- High risk of anchor failure

Why It Matters

- Overload anchor points
- Increase failure risk
- Compromise entire system safety

Simple Rule to Remember

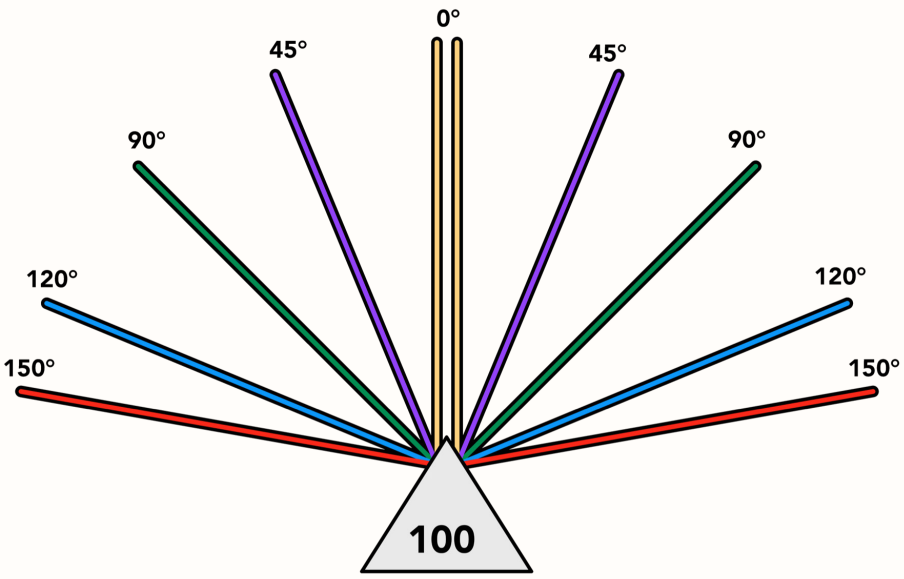
- Smaller angle = Safer system
- Larger angle = Higher force on anchors

Key Takeaway

- Keep angles ≤ 90° whenever possible
- Never approach or exceed 120°

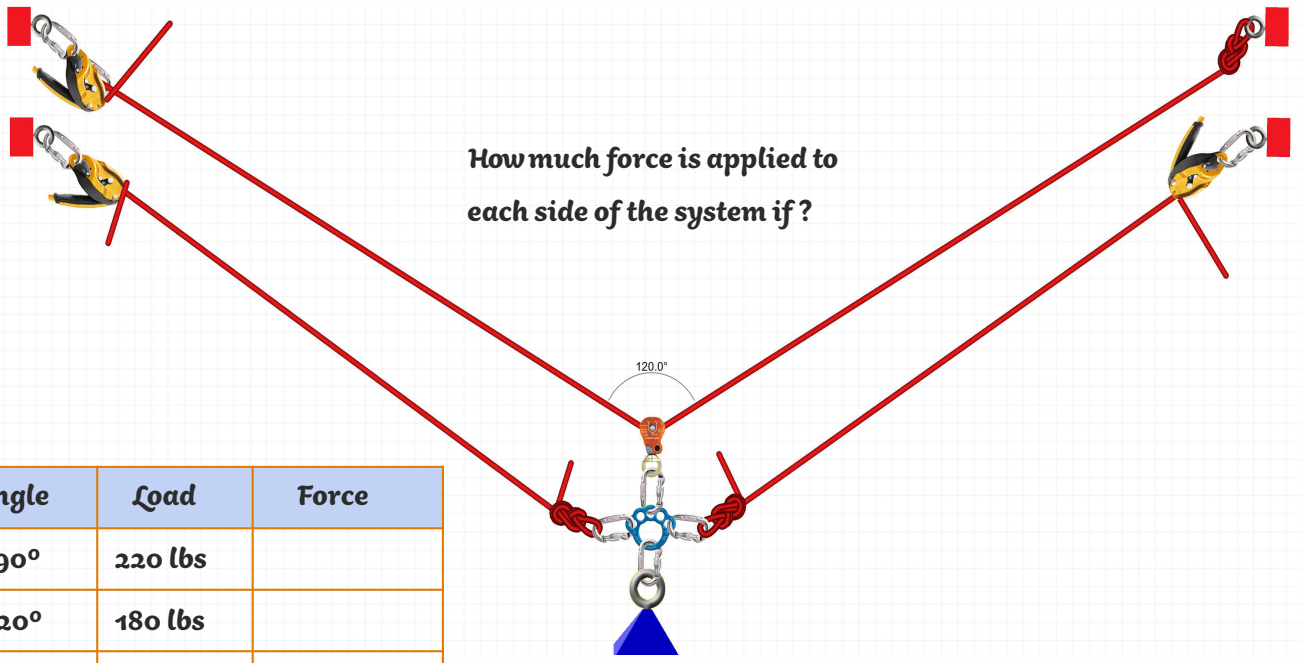
Proper anchor angle control is essential for safe load distribution and system integrity

Included Angles



Angle	Force
0°	50%
45°	57%
90°	71%
120°	100%
150°	200%
160°	300%

Included Angles



Angle	Load	Force
90°	220 lbs	
120°	180 lbs	
160°	220 lbs	

Label the SWL of each piece of equipment, including anchors

WLL vs MBS – Understanding Load Limits

Definitions

Minimum Breaking Strength (MBS)

- The maximum load a component can break before failure
- Determined under controlled test conditions
- Represents the absolute limit (failure point)

Working Load Limit (WLL)

- The maximum safe load during normal use
- Based on a safety factor applied to MBS
- Ensures a margin to prevent failure

General Safety Guidelines

- Solid Metal Components (Carabiners, connectors, anchor hardware)

WLL ≈ 20% (1/5) of MBS

- Fabric & Mechanical Components

(Ropes, slings, harnesses, descenders, ascenders – e.g., ID, Croll)

WLL ≈ 10% (1/10) of MBS

Quick Comparison

MBS

- Failure load
- Not for actual use
- Safety limit reference

WLL

- Safe working load
- Used in real operations
- Includes safety factor

Important Reminder

- Never load equipment near its MBS
- Always operate within WLL limits
- Consider additional forces (shock load, angles, dynamic movement)

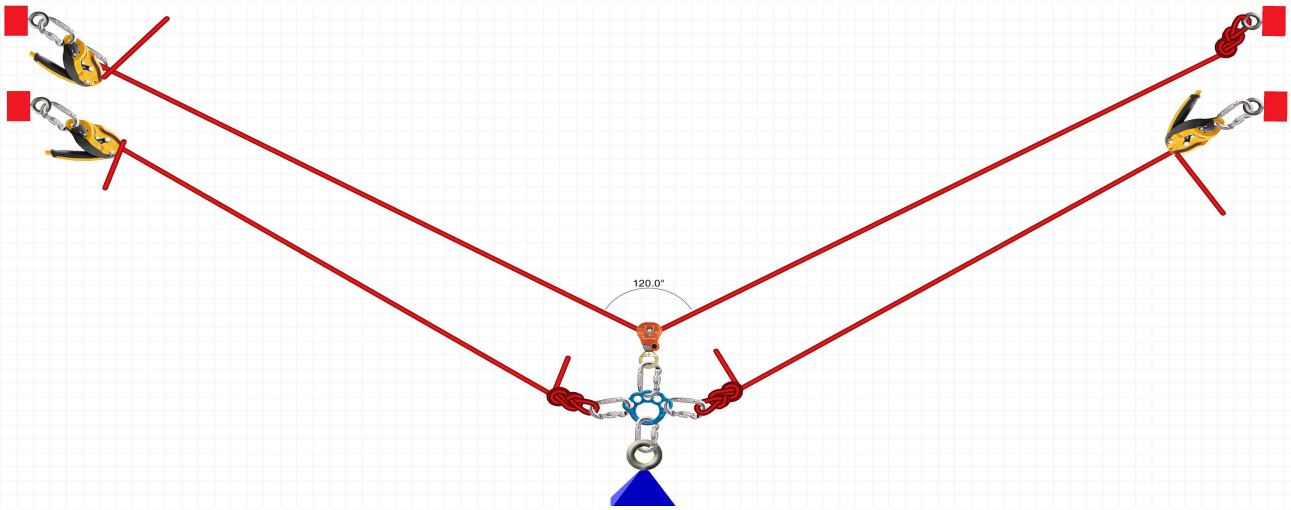
Key Takeaway

MBS = Breaking point

WLL = Safe working limit

Safe systems rely on staying well below failure limits using proper safety factors

Included Angles

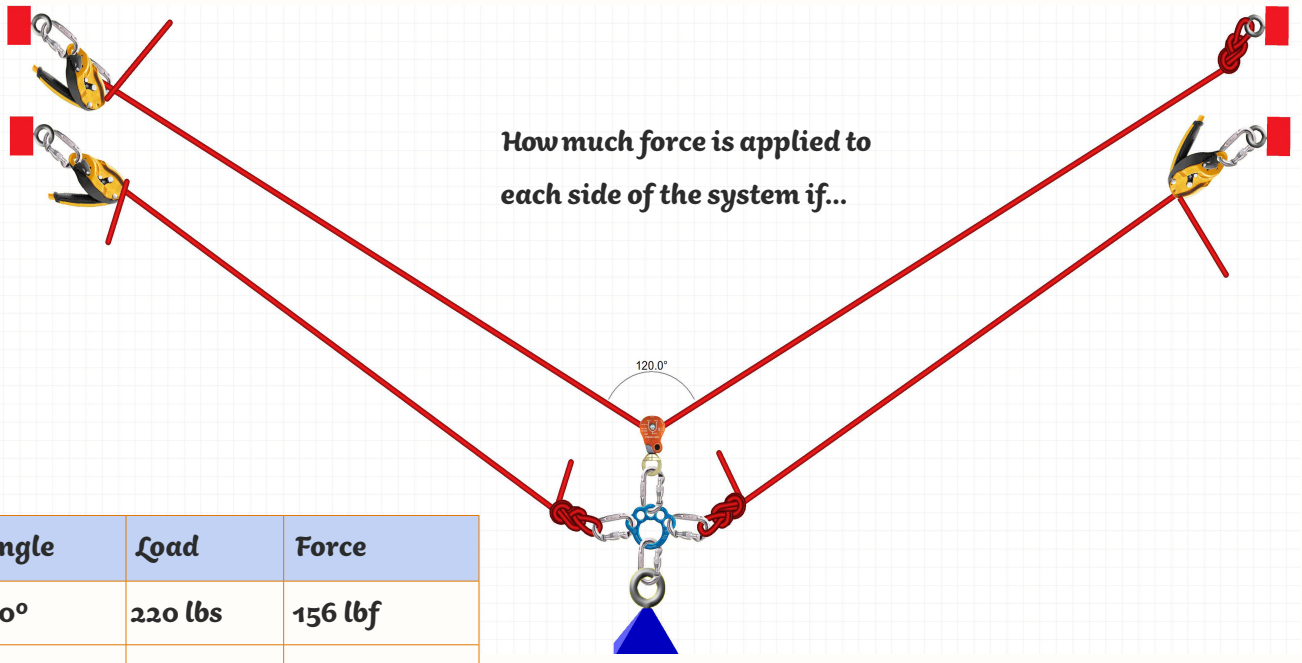


Equipment	MBS	SWL	Equipment	MBS	SWL
Anchor	5000 lbf	1000 lbf	Pulley (Body)	23 kN	1034 lbf
Carabiner	27 kN	1213 lbf	Pulley (shieve)	25 kN	1124 lbf
Rope (with knot)	6000 lbf	420	I'D S	14 kN	330
Rigging plate Paw	36 kN	lbf 1618 lbf	1 kN = 224.8 lbf 1 kg = 2.2 lbs		lbf

SWL of each piece of equipment, including anchors

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Included Angles



Angle	Load	Force
90°	220 lbs	156 lbf
120°	180 lbs	180 lbf
160°	220 lbs	360 lbf

Label the SWL of each piece of equipment, including anchors

Practical Exercises — Day3

LEVEL 1	LEVEL 2	LEVEL 3
Repeat Day2 activities	Rescue Through Knots (S)	Rescue Through Knots (I)
Negotiate Edge	Rescue from Horizontal Aid	Rescue Past Deviation (S - Single) (I — Double)
Rope and Sling Protection	Rescue from Ascent (I)	Mid Transfer Rescue
Anchor Selection	Single Deviation Rescue (I)	Rescue Through Re-anchor
Simple Structural Anchor	Rope to Rope Transfer	Rescue from Short Link Aid
Load Sharing Anchors (< 1m)	Rescue from Fall Arrest (I)	
Hauling and Lowering		

Time to Climb!

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Training Day 4

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Practice and Catch-up

Practice & Performance Guidance

During Training

- Practice and refine your rope access techniques
- Manage your energy — do not overexert yourself
- Prepare for continued performance tomorrow

Learning Actions

- Observe other participants:

5. Learn from different techniques

- Use downtime effectively:

7. Practice knots
8. Practice rigging (even at ground level)

Skill Development

- Focus on challenging techniques
- Avoid over-focusing on one task
- Build overall competence across all skills

Personal Readiness

- Get adequate rest
- Rehydrate after training
- Maintain physical readiness for the next session

Key Takeaway

Improvement depends on:

- Practice
- Observation
- Recovery

Consistent effort and proper recovery lead to better performance and safer work at height

Time to Climb!

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